

sure and an increase in stress and vasoconstriction leading to increased blood pressure as well as increased incidence of coronary artery disease. Acoustic overexposure can lead to temporary or permanent loss of hearing.

Noise can also have a detrimental effect on animals by causing stress, increasing the risk of death by changing the delicate balance in predator/prey detection and avoidance, and interfering with their use of sounds in communication, especially in relation to reproduction and navigation.

Mitigation and control

The roadway noise can be mitigated by using noise barriers, limiting vehicle speeds, altering the roadway surface texture, limiting heavy vehicles, using traffic controls that smooth vehicle flow to reduce braking and acceleration, and tyre design. An important factor in applying these strategies is a computer model for roadway noise that is capable of addressing the local topography, meteorology, traffic operations and hypothetical mitigation.

Costs of building-in mitigation can be modest, provided these solutions are sought in the planning stage of a roadway project. Aircraft noise can be reduced to some extent by design of quieter jet engines.

Although most developed nations have government agencies responsible for protection of the environment, no nation has a single body that regulates noise pollution. In developed parts of the world, different types of noise are managed by agencies responsible for the source of the noise. As the bodies responsible for noise pollution reduction usually view noise as an annoyance rather than a problem, and reducing the noise often hurts the industry financially, little is being done to reduce noise pollution in developed countries.

Active noise control: how is it done?

As all the ways of noise mitigation, control and reduction are proving to be



ineffective due to non-serious attitude of the implementers, there is a need for some more active noise-control techniques.

This can be accomplished with an electronic method of reducing or removing unwanted sound by producing a pressure wave of equal amplitude but opposite sign to the unwanted sound. When the electronically produced inverse wave is added to the original unwanted sound, the result is sound cancellation. This method of noise control is becoming increasingly popular for a variety of uses. It is sometimes considered a miracle 'cure-all' for noise problems.

There are many institutions and companies around the world working on the technology to increase the circumstances where active noise control can be used effectively. In its general form, as applied to the vehicles, the technique involves continuous monitoring of the selected parameters whose effect is to be controlled and producing a cancelling signal (by means of an electronic method). In the

case of active noise cancellation, for example, engine noise is the monitored parameter—selected because of its order-related nature—and the cancellation is provided by a loudspeaker in the passenger cabin at a carefully chosen location.

Active noise control systems need to have phase accuracy of at least a quarter wavelength. After that, the anti-noise starts adding to the background sound, rather than cancelling it out. The size of this bubble of silence is roughly 0.35 metre for a 250Hz tone (roughly middle 'C' on the piano), which has a wavelength of about 1.4 metres. So if somebody is standing next to you and the thing is turned on, they will perceive your quietness

as a series of unpleasant shrieks and squeals. This sensitivity to location means that active noise control systems do not work well above 150 Hz in large spaces where the ear location is not well defined. One way of controlling this space is to use headsets where the position of the speaker and microphone relative to the ear is well defined and the volume is small.

Commercial headsets with active noise-cancellation became available for airplane pilots in the early 1990s and hit the music scene by the end of the decade. Take just about any audio system, turn it up very loud with no signal being played and put your ear close to a speaker in a very quiet room. You will hear some hiss. The hiss you hear in active noise-cancelling headphones is the 'self-noise' or 'noise floor' of the electronics.

The electronics of the active noise-cancellation device involved actively causes noise cancellation in real time. In the case of a single sound frequency, you can add the same frequency 180° out of phase to cancel the sound, but